**Stochastic Processes Semester 1**

**Tutorial 7 – Solutions**

1. (a) i. From Property 1 of PP, the random variables $N(u_1)$, $N(u_1, u_2)$, and $N(u_2, t)$ are mutually independent, and

$$N(u_1) \sim \Pi(\lambda u_1), \quad N(u_1, u_2) \sim \Pi(\lambda(u_2 - u_1)), \quad \text{and} \quad N(u_2, t) \sim \Pi(\lambda(t - u_2)).$$

From that, the random variables $N(u_1, u_2)$ and $\tilde{N} \equiv N(u_1) + N(u_2, t)$ are mutually independent, both have Poisson distribution, and

$$N(u_1, u_2) + \tilde{N} = N(t).$$

Then, from Property Π2,

$$\begin{aligned}\mathbf{P}(N(u_1, u_2) = j \mid N(t) = k) &= \frac{k!}{j!(k-j)!} \left(\frac{\lambda(u_2 - u_1)}{\lambda t}\right)^j \left(1 - \frac{\lambda(u_2 - u_1)}{\lambda t}\right)^{k-j} \\ &= \frac{k!}{j!(k-j)!} \left(\frac{u_2 - u_1}{t}\right)^j \left(1 - \frac{u_2 - u_1}{t}\right)^{k-j}\end{aligned}$$

ii. Using the summation property of covariance and the independence of $N(u_1, u_2)$ and $\tilde{N} = N(0, u_1) + N(u_2, t)$ we have

$$\begin{aligned}\mathbf{Cov}(N(u_1, u_2), N(0, t)) &= \mathbf{Cov}(N(u_1, u_2), \tilde{N}) + \mathbf{Cov}(N(u_1, u_2), N(u_1, u_2)) \\ &= \mathbf{V}ar(N(u_1, u_2)) = \lambda(u_2 - u_1).\end{aligned}$$

(b) We have to find the probability

$$\begin{aligned}\mathbf{P}(N(29; 31) = 0 \mid N(60) = 620) &= \frac{620!}{0!620!} \left(\frac{31 - 29}{60}\right)^0 \left(1 - \frac{31 - 29}{60}\right)^{620} \\ &= \left(\frac{29}{30}\right)^{620} = 7.44^{-10} \approx 0.\end{aligned}$$

Note that the probability does not depend on $\lambda = 10$.

(c) Let $I$ be the interval $(2.29, 2.31)$ and $N_f(I)$, $N_m(I)$ be the number of, respectively, female and male arrivals in this interval, so we want to find

$$\mathbf{P}(N_f(I) = 2 \mid N(I) \leq 3).$$

Then

$$\begin{aligned}\mathbf{P}(N_f(I) = 2 \mid N(I) \leq 3) &= \frac{\mathbf{P}(N_f(I) = 2, N(I) \leq 3)}{\mathbf{P}(N(I) \leq 3)} \\ &= \frac{\mathbf{P}(N_f(I) = 2, N_m(I) \leq 1)}{\mathbf{P}(N(I) \leq 3)} \\ &= \frac{\mathbf{P}(N_f(I) = 2)\mathbf{P}(N_m(I) \leq 1)}{\mathbf{P}(N(I) \leq 3)} \\ &= \frac{(4\lambda/3)^2 e^{-(4\lambda/3)}/2 \left(e^{-(2\lambda/3)} + (2\lambda/3)e^{-(2\lambda/3)}\right)}{\sum_{k=0}^{3} (2\lambda)^k e^{-2\lambda}/k!} \\ &= \frac{(4\lambda/3)^2 \left(1 + (2\lambda/3)\right)}{2\sum_{k=0}^{3} (2\lambda)^k/k!}\end{aligned}$$

and so the probability does depend on the time interval.

2. (a) We require the probability that a Poisson random variable with mean 5 takes the value 0 (or alternatively the probability that an exponential random variable with mean 1 takes a value greater than 5) which is $e^{-5} \approx 0.007$.

   (b) i. By the Splitting Theorem since the process of all customers has rate 1 and customers are independently female or male, female and male customers arrive as *independent* Poisson processes $N_F(\cdot)$ and $N_M(\cdot)$ with respective rates $\alpha$ and $1-\alpha$. Hence the distribution of the time between the arrivals of successive female customers is $\text{Exp}(\alpha)$ (exponential with mean $\alpha^{-1}$).

   ii. In the case $\alpha = 0.2$,

$$\begin{aligned}\mathbf{P}(N_F[0,5] > 2 | N_M[0,5] > 3) &= \mathbf{P}(N_F[0,5] > 2) \quad \text{(independence)} \\ &= 1 - \mathbf{P}(N_F[0,5] \leq 2) \\ &= 0.0803,\end{aligned}$$

   where $N_F[0,5]$ has a Poisson(1) distribution.

   (c) i. The time between Joe's successive conversations is the sum of 3 independent Exp(1) random variables, which is a gamma distribution with shape parameter 3 and scale parameter 1, i.e. with density function $t^2 e^{-t}/2!$

   ii. The required probability is most easily evaluated as the probability that a Poisson random variable with mean 3 takes the value at most 2, which is 0.4232.

3. (a) Regard $T_1, \ldots, T_k$ as the times of occurrence (or locations) of the first point in each of $k$ independent Poisson processes on $\mathbb{R}_+$ with respective rates $\lambda_1, \ldots, \lambda_k$. Then $T$ is the time of occurrence of the first point in the combined process, which, by the Superposition Theorem, is a Poisson process with rate $\lambda$, so that $T$ has the required distribution.

   (b) Alternatively,

$$\begin{aligned}\mathbf{P}(T > t) &= \mathbf{P}(T_j > t,\ j = 1, \ldots, n) \\ &= \prod_{j=1}^{n} \mathbf{P}(T_j > t) \qquad \text{(independence)} \\ &= \prod_{j=1}^{n} \exp(-\lambda_j t) \\ &= \exp(-\lambda t).\end{aligned}$$

4. (a) By the Superposition Theorem total injuries occur as a Poisson process with rate 12, and each is male with probability 5/6 independently of the others. The probability that 8 male injuries occur before 4 female is the probability that, of the first 11 injuries, at least 8 are male. This is the probability that a binomial random variable with parameters 11 and 5/6 takes the value at least 8, which is 0.904.

   (b) i. By the Superposition Theorem as above, we have, for $n \geq 1$,

$$X_n = \begin{cases} X_{n-1} + 1 & \text{with probability } 5/6 \\ X_{n-1} - 1 & \text{with probability } 1/6 \end{cases}$$

   independently of the previous history of the process, and so $(X_n,\ n \geq 0)$ is a simple random walk as required.

   ii. The probability $x$ is the probability that the process $(X_n,\ n \geq 0)$ is ever strictly negative ($X_n \leq -1$ for some $n$). This is the probability of ruin for the process $(X_n,\ n \geq 0)$ but starting at 1 instead of 0. Hence

$$x = \min\left(1, \left(\frac{1/6}{5/6}\right)^1\right) = 1/5.$$

5. (a) This is just the Splitting Theorem.

   (b) i. Let $T$ be the time until a total of $n$ claims have occurred. Then $T$ is the sum of $n$ independent $\text{Exp}(\lambda)$ random variables and so $E(T) = n\lambda^{-1}$, $\text{var}(T) = n\lambda^{-2}$.

   ii. For any interval $I$ of unit length let

$$I_k = \begin{cases} 1 & \text{if a claim of type } k \text{ occurs in } I \\ 0 & \text{otherwise} \end{cases}$$

Then, by the Splitting Theorem, $I_1, \ldots, I_n$ are independent and

$$\mathbf{P}(I_k = 1) = 1 - \exp(-\lambda/n).$$

The number of *different* claims arising in the interval $I$ is given by

$$Y = \sum_{k=1}^{n} I_k.$$

It follows $Y$ is binomial with parameters $n$ and $1 - \exp(-\lambda/n)$. Hence

$$E(Y) = n(1 - \exp(-\lambda/n)); \quad \text{var}(Y) = n\exp(-\lambda/n)(1 - \exp(-\lambda/n)).$$

6. Let $A_1$ and $A_2$ respectively be the events that the 1st and 2d stamps turn up first. Let $T'$ be the time to the first occurrence of a stamp of either type, and let $T$ be the time until at least one example of each type is obtained. Then, by the Superposition Theorem, $E(T') = 1/(\lambda_1 + \lambda_2)$. Further, again by the Superposition Theorem, the event $A_1$ occurs with probability $\lambda_1/(\lambda_1+\lambda_2)$ and then the (conditional) expectation of the further time $T - T'$ required to obtain an example of the 2d is $1/\lambda_2$, while the event $A_2$ occurs with probability $\lambda_2/(\lambda_1+\lambda_2)$ and then the (conditional) expectation of the further time $T - T'$ required to obtain an example of the 1st is $1/\lambda_1$. Thus

$$\begin{aligned} E(T) &= E(T') + E(T - T') \\ &= E(T') + \mathbf{P}(A_1)E(T - T'|A_1) + \mathbf{P}(A_2)E(T - T'|A_2) \\ &= \frac{1}{\lambda_1 + \lambda_2} + \frac{\lambda_1}{\lambda_1 + \lambda_2}\frac{1}{\lambda_2} + \frac{\lambda_2}{\lambda_1 + \lambda_2}\frac{1}{\lambda_1} \\ &= \frac{\lambda_1^2 + \lambda_2^2 + \lambda_1\lambda_2}{(\lambda_1 + \lambda_2)\lambda_1\lambda_2}. \end{aligned}$$

An alternative argument is to let $T_1$ be the time until the first occurrence of the 1st. We have $E(T_1) = 1/\lambda_1$, $E(T - T_1|A_1) = 1/\lambda_2$, $E(T - T_1|A_2) = 0$. Now argue as above, with $T_1$ in place of $T'$.

The quickest argument is to observe that $T$ is the maximum of independent $\text{Exp}(\lambda_1)$ and $\text{Exp}(\lambda_2)$ random variables, say $T_1$ and $T_2$. We know that the minimum of independent $\text{Exp}(\lambda_1)$ and $\text{Exp}(\lambda_2)$ random variables has exponential distribution with rate $\lambda_1 + \lambda_2$. Since we can write the maximum of two random variables as their sum minus their minimum

$$T = \max(T_1, T_2) = T_1 + T_2 - \min(T_1, T_2),$$

it follows that

$$E(T) = E(T_1) + E(T_2) - E(\min(T_1, T_2)) = \frac{1}{\lambda_1} + \frac{1}{\lambda_2} - \frac{1}{\lambda_1 + \lambda_2} = \frac{\lambda_1^2 + \lambda_2^2 + \lambda_1\lambda_2}{(\lambda_1 + \lambda_2)\lambda_1\lambda_2}.$$